

Genetics Quiz 2

- Due No due date
- Points 20
- Questions 20
- Time Limit 12 Minutes
- Allowed Attempts Unlimited

Take the Quiz Again

Attempt History

Attempt	Time	Score
LATEST	Attempt 1	11 minutes
		20 out of 20

Score for this attempt: 20 out of 20

Submitted Dec 27 at 7:17pm

This attempt took 11 minutes.

Correct answer

⋮

Question 1

1 / 1 pts

In a population of 1,000 mice, the coat color is determined by a single gene with two alleles: a dominant allele (B) for black fur and a recessive allele (b) for white fur. Researchers observe that 160 mice in the population have white fur. Assuming the population is in Hardy-Weinberg equilibrium, calculate the frequency of the dominant allele (B). Given the above scenario, what is the frequency of the dominant allele (B) in the mouse population?

- ☐ 0.64
- ☐ 0.36
- ☐ 0.40
- ☒ 0.60

Correct answer

⋮

Question 2

1 / 1 pts

The Hardy-Weinberg equilibrium provides a mathematical framework to study genetic variation in populations. It assumes that allele and genotype frequencies remain constant over time in the absence of evolutionary forces. Which of the following would disrupt a population's Hardy-Weinberg equilibrium?

- ☒ Migration of individuals into or out of the population

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- ☒ 50%
- ☐ 75%
- ☐ 100%

Correct answer

⋮

Question 6

1 / 1 pts

What are the two major kinds of molecules that direct the synthesis of proteins in eukaryotic cells?

- ☒ DNA and RNA
- ☐ DNA and ATP
- ☐ RNA and ATP
- ☐ DNA and ADP

Correct answer

⋮

Question 7

1 / 1 pts

What procedure is utilized in determining the presence of genetic disorder in an unborn child?

- ☐ In vitro Fertilization
- ☐ Gene manipulation
- ☐ DNA test
- ☒ Amniocentesis

Correct answer

⋮

Question 8

1 / 1 pts

Mutation can take place in any part of the body at any time. However, it is most likely to spread through a population if it is present in what of the following?

- ☒ Spermatozoa
- ☐ Neurons
- ☐ Epithelial cells
- ☐ Bone cells

Correct answer

⋮

Question 9

1 / 1 pts

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- ☐ Random mating within the population
- ☐ No mutations occurring in the gene pool
- ☐ A large population size with minimal genetic drift

Correct answer

⋮

Question 3

1 / 1 pts

Chargaff's rules are fundamental to understanding DNA structure. According to these rules, in any DNA molecule, the amount of adenine (A) equals the amount of thymine (T), and the amount of cytosine (C) equals the amount of guanine (G). This base-pairing rule forms the basis for the double-helix structure of DNA, as described by Watson and Crick. If a DNA molecule contains 30% adenine (A), what percentage of the molecule is guanine (G)?

- ☒ 20%
- ☐ 30%
- ☐ 40%
- ☐ 60%

Correct answer

⋮

Question 4

1 / 1 pts

Hypophosphatemia, a condition in which phosphate levels in blood is low is inherited as an X-linked dominant trait. If a man with this condition marries a healthy woman, which of their offspring would most likely inherit the disease?

- ☐ All males
- ☒ All females
- ☐ Half of the males
- ☐ Half of the females

Correct answer

⋮

Question 5

1 / 1 pts

The mutation of certain genes is lethal in both homozygous and heterozygous recessives. For instance, Manx cats carry the mutated allele ML. Homozygous cats M^LM^L dies before birth while heterozygous M^LM have a spine that developed abnormally causing tail lessness. If two M^LM cats' mate, what is the probability that any of their surviving kittens will be heterozygous?

- ☐ 25%

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A man with curly hair married a woman who also has curly hair. Of their four children, three have curly hair, while one has straight hair. What are the possible genotypes of the parents?

- ☐ The father is CC and the mother is Cc.
- ☐ The father is Cc and the mother cc.
- ☐ Both parents are CC.
- ☒ Both parents are Cc.

Correct answer

⋮

Question 10

1 / 1 pts

Which of the following application of genetic engineering poses a threat to the society?

- ☐ Transgenice goats produce proteins used to treat cystic fibrosis, heart attacks, blood clotting disorders, and even nerve gas exposure.
- ☐ Transgenic crop plants are being developed for regions that are affected by severe droughts.
- ☒ Increasing demand on organ transplant with short supply of human donors.
- ☐ Mice are used to study human diseases and their potential cures without experimenting on humans.

Correct answer

⋮

Question 11

1 / 1 pts

The last 23rd pair of chromosomes determines the gender of the human. For instance, Turner syndrome involves only a single sex chromosome (X). Which of the following does NOT characterize this error in sex chromosome?

- ☐ Short Stature
- ☐ Sterility
- ☐ Wedded neck
- ☒ Small testes

Correct answer

⋮

Question 12

1 / 1 pts

Genetic engineering offers numerous advantages and disadvantages in various fields. In one application, a gene from a species of one genotype is introduced to the genome of another species. What do you call such an organism?

- ☐ Invasive Species
- ☐ Mutated Species

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- ☒ Transgenic Species

☐ Keystone Species

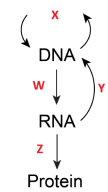
Correct answer



Question 13

1 / 1 pts

The schematic diagram shown above represents the modern central dogma of molecular genetics. What process is represented by z?



- ☒ Translation
- ☐ Replication
- ☐ Transcription
- ☐ Reverse Transcription

Correct answer



Question 14

1 / 1 pts

Albinism happens when a person has no visible skin pigment due to a mutation of genes that produces tyrosinase, a protein that is accountable for normal skin pigment. What type of non-Mendelian law is given this situation?

- ☐ Pleiotropy
- ☒ Epistasis
- ☐ Multiple Alleles
- ☐ Codominance

Correct answer



Question 15

1 / 1 pts

Which of the following statements about mutations is NOT correct?

- ☐ Mutations bring about variation for natural selection to act on
- ☐ Mutations arise spontaneously, rarely and at random
- ☒ Most mutations produce alleles which are dominant

- ☐ The rate of mutations can be increased by radiation and other artificial means

Correct answer



Question 16

1 / 1 pts

Marfan Syndrome results in several symptoms such as unusually tall height, thin finger and toes, lens dislocation and heart problems. These symptoms do not seem related, but as it turns out, they can all be traced back to the mutation of single gene. What type of non Mendelian law is given on this situation?

- ☐ Lethal alleles
- ☐ Codominance
- ☒ Pleiotropy
- ☐ Multiple Alleles

Correct answer



Question 17

1 / 1 pts

In the production of a genetically modified organism such as bacteria producing human insulin, what done first?

- ☐ Cut open the plasmid by restriction enzymes
- ☐ Insert the insulin gene into the plasmid vector using DNA ligase
- ☐ Introduce a recombinant DNA into a bacterial cell
- ☒ Extract human insulin from pancreases to isolate an insulin-producing gene

Correct answer



Question 18

1 / 1 pts

A team of developmental biologists is studying the genetic mechanisms that determine the body axis formation in animals. They observe that mutations in a specific set of genes lead to abnormalities in head and tail development, resulting in disrupted body segmentation. These genes play a crucial role in establishing the anterior-posterior axis during embryonic development. Which of the following gene families is primarily responsible for regulating the proper formation of the head and tail in animal development?

- ☒ HOX
- ☐ TP53
- ☐ OCT 3/4
- ☐ NANOG

Correct answer



Question 19

1 / 1 pts

Scientists use genetic engineering to modify organisms for various purposes, such as improving crop resistance, producing medicines, and enhancing beneficial traits. However, there are concerns about how these genetically modified organisms (GMOs) might affect natural ecosystems if they are introduced into the wild. What do you call a species that has been genetically engineered to contain genes from another species?

- ☐ Genetically edited species
- ☐ Endemic species
- ☒ Transgenic species
- ☐ Genetically mutant species

Correct answer



Question 20

1 / 1 pts

In a genetic counseling session, Dr. Smith is reviewing the inheritance patterns of hair texture with a couple. Both individuals have curly hair and are considering the genetics behind their children's traits. They have four children: three who also have curly hair and one who has straight hair. Dr. Smith is trying to determine the likely genotypes of the parents based on these observations. What are the possible genotypes of the parents if curly hair is dominant (A) and straight hair is recessive (a) ?

- ☐ The father is heterozygous (Aa) and the mother is homozygous recessive (aa).
- ☒ Both parents are heterozygous (Aa).
- ☐ The father is homozygous dominant (AA) and the mother is heterozygous (Aa).
- ☐ Both parents are homozygous dominant (AA).

Quiz Score: 20 out of 20